

B. Tech. 1st Semester (Mid Semester-I) Examination-2023

BIT Sindri, Dhanbad

Subject: Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSC 01

Answer any five questions.

1. (a) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ then $A^2 - 4A + 3I$ is equal to [CO 1]

(i) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(iv) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ [CO 1]

- (b) The vectors $u=(6,2,3,4)$, $v=(0,5,-3,1)$, $w=(0,0,7,-2)$ are

- (i) Linearly dependent
(iii) Insufficient data

- (ii) Linearly Independent
(iv) None of these

- (c) The Eigen values of the matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ 8 & 4 & 0 \\ 6 & 2 & 5 \end{bmatrix}$ are [CO 1]

(i) 3,4,5

(ii) 1,1,1

(iii) 2,3,6

(iv) None

- (d) The number of asymptotes of the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ are [CO 2]

(i) 1

(ii) 2

(iii) 4

(iv) None

2. Convert the matrix $A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$ into normal form and hence find its rank. [CO 1]

3. Determine the characteristic roots and characteristics vectors of the $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. [CO 1]

4. Investigate for what values of a and b, the equations $x+2y+3z=6$; $x+3y+5z=9$, $2x+5y+az=b$ have (a) no solution (b) a unique solution (c) Infinite solution. [CO 1]

5. Find the dimension of the subspace of R_4 spanned by the set $\{(1\ 0\ 0\ 0), (0\ 1\ 0\ 0), (1\ 2\ 0\ 1), (0\ 0\ 0\ 1)\}$. Hence find its basis. [CO 1]

6. Expand $\sin x$ in powers of $x - \frac{\pi}{2}$. Hence evaluate $\sin 91^\circ$ correct to four decimal places. [CO 2]

7. Find the asymptotes of the curve $y^3 - x^2y - xy^2 + x^3 - y^2 + x^2 = 0$. [CO 2]

B. Tech. 1st Semester (Mid Semester-II) Examination-2023

BIT Sindri, Dhanbad

Subject: Engg. Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSM 01

Answer any five questions.

1. (a) If $x = r \cos \theta$, $y = r \sin \theta$ then $\frac{\partial(r, \theta)}{\partial(x, y)}$ is equal to [CO 3]

(i) r (ii) $-r$ (iii) $\frac{1}{r}$ (iv) $-\frac{1}{r}$

- (b) If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ and $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$, then (a, b) is a maximum point of $f(x, y)$ when [CO 3]

(i) $rt - s^2 > 0, r > 0$ (ii) $rt - s^2 < 0, r > 0$ (iii) $rt - s^2 > 0, r < 0$ (iv) None of these

- (c) The value of $\int_1^2 \int_2^3 \int_3^4 dx dy dz$ is [CO 4]

(i) 0 (ii) 1 (iii) 2 (iv) 3

- (d) The double integral $\int_0^{\frac{\pi}{2}} \int_0^1 r \sin \theta dr d\theta$ can be evaluated as [CO 4]

(i) $\frac{1}{2}$ (ii) 1 (iii) $\frac{1}{3}$ (iv) 0

2. If $u = x + 3y^2 - z^3$, $v = 4x^2 yz$, $w = 2z^2 - xy$ then evaluate $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at $(1, -1, 0)$. [CO 3]

3. Expand $x^2 y + 3y - 2$ in powers of $(x-1)$ and $(y-2)$ using Taylor's theorem. [CO 3]

4. Discuss the maxima and minima of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. [CO 3]

5. Evaluate the integral $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ using change of order of integration. [CO 4]

6. Prove that $\int_0^1 x^m (\log x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$ where n is a positive integer and $m > -1$. [CO 4]

7. Using double integration, find the area lying inside the circle $r = a \sin \theta$ and outside the cardioid $r = a(1 - \cos \theta)$. [CO 4]

B. Tech. 1st Semester (Supplementary Mid Semester) Examination-2023

BIT Sindri, Dhanbad

Subject: Engg. Mathematics-I (All Branches)

Full Marks : 4x5=20

Time : 1 hr 30Mins

Subject Code : BSM 01

Answer any five questions.

- ✓ 1. (a) The product of any matrix with its transpose is [CO 2]
(i) symmetric (ii) skew-symmetric
(iii) Orthogonal (iv) None of these
- (b) If $f_x = 3x + 2y$ and $f_y = 2x + 3y$ then $f(x,y)$ has [CO 3]
(i) minimum at (0,0) (ii) maximum at (0,0)
(iii) saddle point at (0,0) (iv) None of these
- (c) The value of $\iint_R x^2 y dx dy$, where $R : \{0 \leq x \leq 2; 0 \leq y \leq 1\}$ is [CO 4]
(i) 2/3 (ii) 1 (iii) 4/3 (iv) 0
- (d) The vector field $\vec{F} = y\vec{i} + x\vec{j}$ is [CO 5]
(i) irrotational (ii) solenoidal
(iii) both irrotational and solenoidal (iv) None of these
- ✓ 2. Determine whether (1,1,1,1), (1,2,3,2), (2,5,6,4), (2,6,8,5) form a basis of R^4 . If not find the dimension of the subspace they span. [CO 1]
3. Prove that
$$\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix} = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right).$$
 [CO 2]
- ✓ 4. If $f = \log r$ and $r^2 = x^2 + y^2 + z^2$, then show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{1}{r^2}$. [CO 3]
5. Prove that the product of three positive numbers of constant sum is maximum when they are equal. [CO 3]
- ✓ 6. Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) dx dy$ by changing to polar co-ordinates. [CO 4]
- ✓ 7. Find the directional derivatives of $f(x,y,z) = x^2 yz + 4xz^2$ at the point (1,2,-1) in the direction of the vector $2\vec{i} - \vec{j} - 2\vec{k}$. [CO 5]
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Department of Electrical Engineering
BIT Sindri, Dhanbad-828123
B. Tech 1st Semester Second Mid Semester Examination-2023

Subject: Basics of Electrical Engineering (BEE)

Subject Code: ESEE1

Section: G, H & I

Full Marks= 4X5= 20

Branch: EE, ECE, CSE, IT

Time: 1 Hour 30 Minutes

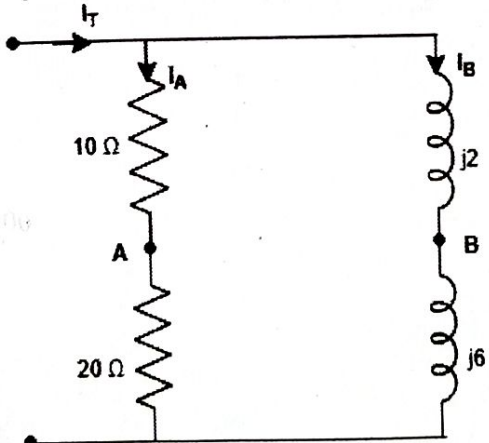
Instructions

A. Answer **Any Five** questions

B. Write in your own words.

B. Each Question carries Equal marks.

C. Assume missing data suitably.

Q.No.	Description	Marks	CO's
1 (a).	A supply of 400 V, 50 Hz is applied to a series R-C circuit. Calculate the value of C if the power absorbed by the resistor be 500 W at 150 V. Also, determine the energy stored in capacitor?	2	CO2
(b).	When a resistor and an inductor are connected in series to 240 V, 50 Hz a.c mains, a current of 3 A flows lagging 37° behind the supply voltage, while the voltage across the inductor 171 V. Compute the resistance of the resistor, resistance and reactance of the inductor.	2	
2 (a).	A resistance of 12Ω , inductance of 0.1 H, and a capacitance of $100 \mu\text{F}$ are connected in series across 220 V, 100 Hz ac supply. Determine the current and its power factor, power consumed.	2	CO2
(b).	A circuit with an impedance $Z = 3 + j4 \Omega$ and an applied voltage (phasor) $V = 100 \angle 30^\circ \text{ V}$. Determine the power triangle?	2	
3 (a).	Derive an expression for the average power in a 1- ϕ series R-L circuit.	2	CO2
(b).	The total current entering to the parallel circuit is given by $I_T = 18 \angle 45^\circ \text{ A}$. Compute the potential difference between points A & B. 	2	

4 (a).	A series R-L-C circuit consisting of a resistor of $20\ \Omega$, inductance of 0.2 H and a capacitance of $150\ \mu\text{F}$ is connected across 230 V , 50 Hz supply. Determine (i) impedance (ii) current (iii) frequency of supply to be adjusted to make power factor unity.	2	CO2
(b).	A circuit consists of a $115\ \Omega$ resistor, a $0.024\ \mu\text{F}$ capacitor and a coil of inductance L . If the resonant frequency of the circuit is 1 KHz . Compute inductance and bandwidth?	2	
5 (a).	Explain the impedance triangle of inductive load & capacitive load with suitable phasor diagram?	2	CO2
(b).	A voltage $v = 100 \sin(\omega t + 30^\circ)$ and current flowing in the circuit is $i = 20 \sin(\omega t + 60^\circ)$. Determine the impedance, resistance, reactance and power factor.	2	
6 (a).	Derive the relationship between phase and line quantities of voltages and currents in a 3- ϕ star (Y) connected circuit. Draw the phasor diagram?	2	CO2
(b).	Three equal impedances each having a resistance of $25\ \Omega$ and reactance of $40\ \Omega$ are connected in star (Y) to a 400 V , 3- ϕ , 50 Hz system. Determine (i) line current (ii) power factor (iii) power consumed.	2	
7 (a).	Derive the e.m.f. equation of 1- ϕ transformer?	2	CO3
(b).	A 25 KVA ideal transformer has 500 turns on the primary and 40 turns on the secondary windings. The primary is connected to 3 KV , 50 Hz mains supply. Determine (i) primary current (ii) secondary current (iii) secondary e.m.f (iv) Maximum flux in the core (No-load current can be neglected)	2	

Department of Electrical Engineering
BIT Sindri, Dhanbad-828123
B. Tech 1st Semester Mid Semester Examination 2023

Subject: Basics of Electrical Engineering (BEE)

Section: G, H & I

Full Marks= 4X5= 20

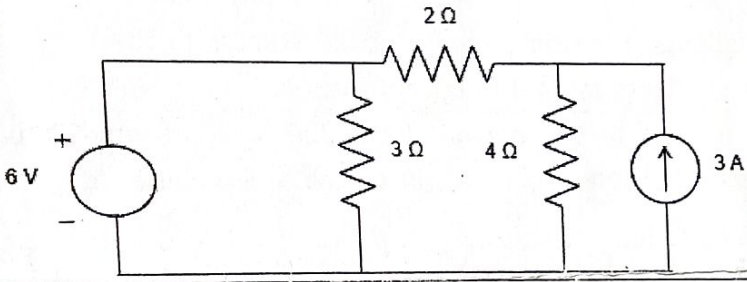
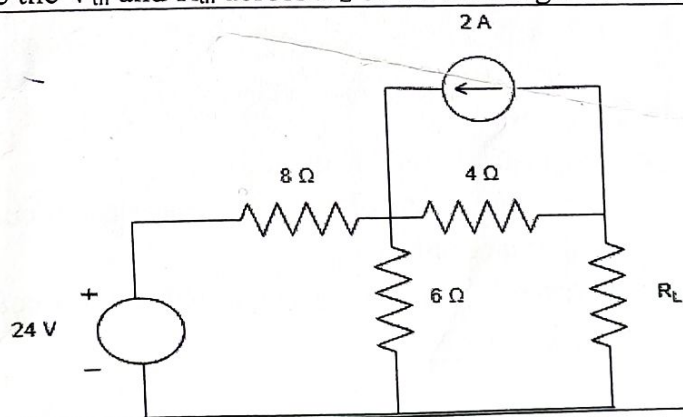
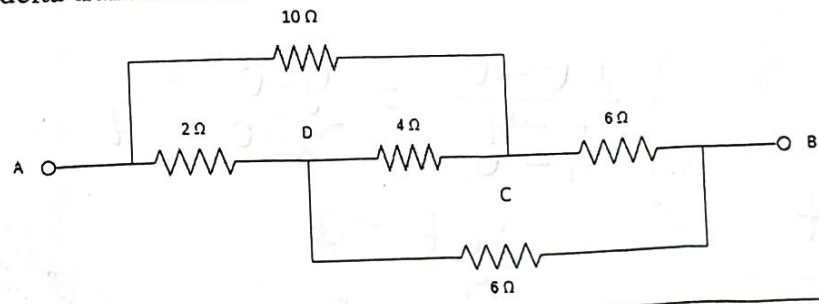
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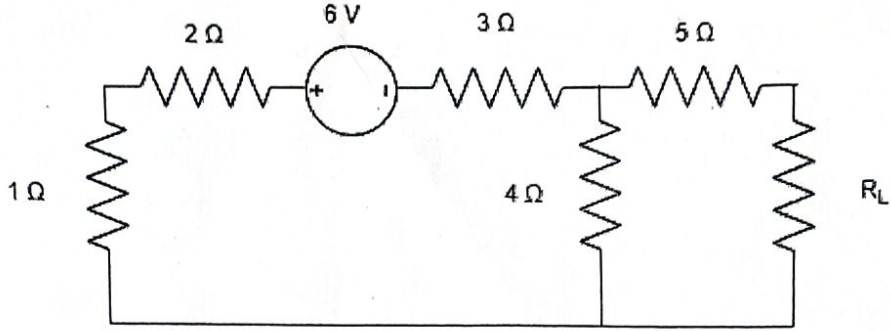
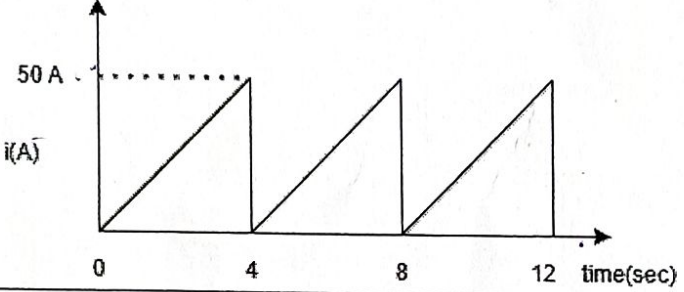
Branch: EE, ECE, CS&IT

Time: 1 Hour 30 Minutes

Instructions

- A. Answer **Any Five** questions
- B. Write in your own words.
- B. Each Question carries Equal marks.
- C. Assume missing data suitably.

Q.No.	Description	Marks	CO's
1 (a).	State and explain the Superposition theorem?	1	CO1
(b).	Determine the current in different resistors using superposition theorem in the circuit given below.	3	
			
2 (a).	State Thevenin's theorem?	1	CO1
(b).	Determine the V_{th} and R_{th} across R_L of the circuit given below	3	
			
3 (a).	Explain KVL and KCL?	2	CO1
(b).	Compute the equivalent resistance between terminals A & B using star-delta transformation	2	
			

4 (a).	State Maximum power transfer theorem?	1	
(b).	Determine the value of R_L which will absorb the maximum power from the circuit shown below. Also, calculate the value of maximum power?	3	CO1
			
5 (a).	Calculate the r.m.s current value of the wave given below?	2	CO1
			
(b).	An alternating voltage $v(t) = 141.4 \sin 314t$. Determine (i) frequency (ii) average value (iii) the time taken for the voltage to reach 100 V for the first time after passing through zero value.	2	
6 (a).	A 100 V, 60 W bulb to be operated from 220 V, AC supply. What resistance must be connected in series with the bulb to glow normally?	2	
(b).	A 230 V, 50 Hz voltage is supplied to an inductor of an inductance 0.2 Write the voltage and current equation. Also determine the maximum energy stored in the inductor?	2	CO1
7	Write short notes (any two) (i) Norton's theorem (ii) Star-delta transformation (iii) Average value of an alternating quantity (iv) Compute current and power of the single-phase a.c circuit containing inductance only (v) Compute current and power of the single-phase a.c circuit containing capacitance only	2x2	CO1

Department of Electrical Engineering
BIT Sindri, Dhanbad-828123
B. Tech 1st Semester Supplementary Mid Semester Examination-2023

Subject: Basics of Electrical Engineering (BEE)

Subject Code: ESEE1

Section: G, H & I

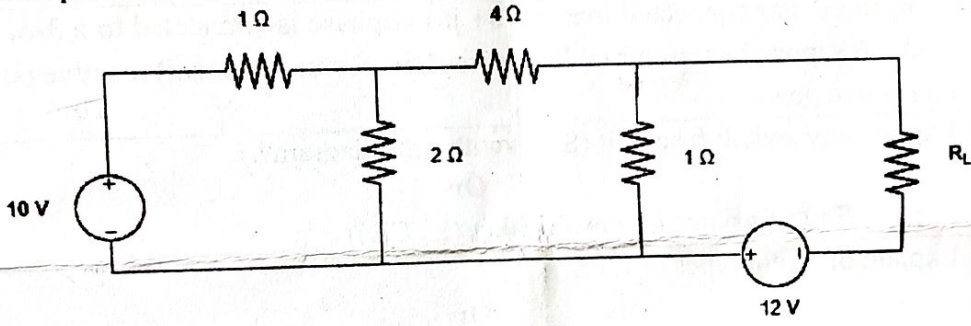
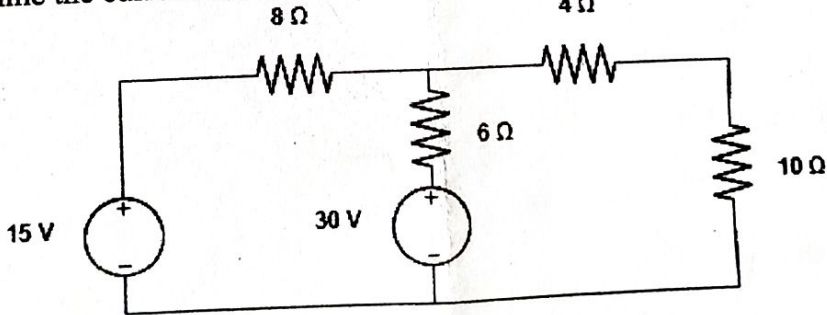
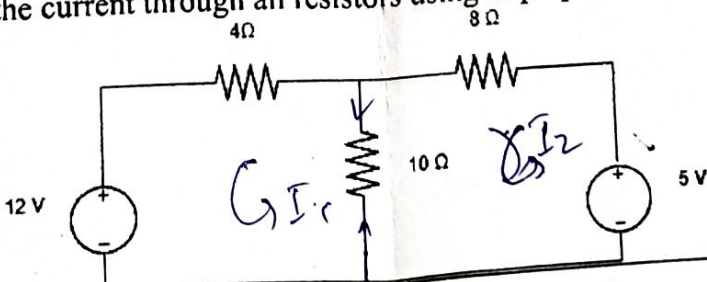
Branch: EE, ECE, CSE, IT

Full Marks= 4X5= 20

Time: 1 Hour 30 Minutes

Instructions

- A. Answer Any Five questions
- B. Write in your own words.
- B. Each Question carries Equal marks.
- C. Assume missing data suitably.

Q.No.	Description	Marks	CO's
1 (a).	State & Explain Superposition theorem with example?	2	CO1
(b).	Determine the value of load resistance R_L such that maximum power transfer takes place. Also, calculate the maximum power? 	2	
2 (a).	State Thevenin's theorem and illustrate with help of suitable example.	2	CO1
(b).	Determine the current in 10 Ω resistor by using Norton's theorem? 	2	
3 (a).	Determine the current through all resistors using Superposition theorem? 	2	CO1

(b).	Three resistances of $20\ \Omega$, $25\ \Omega$ and $30\ \Omega$ are connected in delta. Calculate the corresponding resistances in an equivalent star connection.	2	
4 (a).	The equation of an alternating current $i = 42.42 \sin 628t$. Determine (i) frequency (ii) r.m.s value (iii) average value (iv) form factor	2	CO1
(b).	A coil having a resistance $7\ \Omega$ and an inductance of $31.8\ \text{mH}$ is connected across $230\ \text{V}$, $50\ \text{Hz}$ main supply. Determine (i) circuit current (ii) phase angle (iii) power factor (iv) power consumed	2	
5 (a).	A voltage of $200\ \text{V}$ is applied to a series circuit consisting of a resistor, an inductor and a capacitor. The respective voltages across these components are $170\ \text{V}$, $150\ \text{V}$, $100\ \text{V}$ and the current is $4\ \text{A}$. Determine (i) resistance (ii) impedance (iii) inductive reactance (iv) capacitive reactance.	2	CO1
(b).	A circuit consisting a resistance of $4\ \Omega$, an inductance of $0.5\ \text{H}$ and a variable capacitance are connected in series across $100\ \text{V}$, $50\ \text{Hz}$ supply. Calculate (i) the capacitance at resonance (ii) voltage across inductance and capacitance (iii) Q- factor of the circuit.	2	
6 (a).	What are the different types of cables? Explain briefly?	2	CO1, CO5
(b).	A balanced star connected load of $(8 + j6)\ \Omega/\text{phase}$ is connected to a $3\text{-}\phi$, $230\ \text{V}$ mains supply. Determine (i) line current (ii) power factor (iii) reactive power (iv) active power.	2	
7 (a).	Explain any switch fuse unit (SFU) with neat diagram? Or Explain Earth Leakage Circuit Breaker (ELCB)?	2	CO5
(b).	Explain different types of UPS? Or Explain Earthing and its importance?	2	



DEPARTMENT OF CSE & IT
BIT SINDRI DHANBAD
First Mid-Semester Examination-2023

Subject – Programming for Problem Solving
Semester – 1ST (CSE/IT/EE/ECE)

Time -1.5 Hour
Subject Code – ESPP1

Note: Read all the questions carefully, do as directed. Attempt any 5 questions.

1. Write a program which takes two integer values from user, a number and a digit between 0 to 9 and find whether the digit is present in the number or not.
- [CO1] [4 Marks]

2. Write a program to find the difference of the sum of all even numbers and the sum of all odd numbers present in an array.

[CO1, CO3] [4 Marks]

3. Write a program to print the below pattern where the number of lines to print will be taken from user.

1
123
12345
1234567
123456789

[CO1, CO3] [4 Marks]

4. Draw flowchart and write algorithm to compute the factorial of a number.

[CO2] [4 Marks]

5. Write a program to find the second smallest number in an array.

[CO1, CO3] [4 Marks]

6. What is operator? Write any five different types of operators in C with their precedence and associativity.

[CO1] [4 Marks]

7. Convert the below decimal numbers into their equivalent number of the given base.

- a. $(510)_{10}$ = $(\quad)_{16}$
b. $(-23)_{10}$ = $(\quad)_{2}$
c. $(49)_{10}$ = $(\quad)_{2}$
d. $(214)_{10}$ = $(\quad)_{8}$

[CO1] [4 Marks]

*****END*****

DEPARTMENT OF CSE & IT
BIT SINDRI DHANBAD
Supplementary Mid-Semester Examination-2023

Subject – Programming for Problem Solving
Semester – 1ST (CSE/IT/EE/ECE)

Time -1.5 Hour
Subject Code – ESPP1

Note: Read all the questions carefully, do as directed. Attempt any 5 questions.

1. Consider the data
78, 49, 32, 53, 88, 64, 30, 52
What will be output after completion of 4th pass?
(a) Using insertion sort
(b) Using bubble sort

[CO1] [4 Marks]
- ✓ 2. Explain the different types of loops in C with examples.

[CO3] [4 Marks]
3. Draw a flow chart and write a C program to compute the GCD of two numbers.

[CO2] [4 Marks]
4. Write a C program to count the number of vowels in a string given by the user.

[CO3] [4 Marks]
- ✓ 5. Write-Output
(a)

```
for ( i = 1; i < 6; i++){  
    for ( j = 0; j < i; j++){  
        printf("%d", (j*2)+1);  
    }  
    printf("\n"); }
```


(b)

```
for ( i = 1; i < 6; i++){  
    for ( j = 0; j < i; j++){  
        printf("%d", j%2);  
    }  
    printf("\n"); }
```

[CO3] [4 Marks]
- ✓ 6. Fill in the blanks for the following
(a) The _____ function is used to copy the string in C.
(b) The _____ operator is used for performing logical OR in C.
(c) The _____ function is used to find the length of a string in C.
(d) The _____ statement is used to skip the current iteration of a loop in C.

[CO3,CO4] [4 Marks]
7. (a) Write rules for naming the variable in C.
(b) What is the significance of function? Suppose the concept of function is eliminated in any programming language. What will be the demerit of that?

[CO1,CO4] [4 Marks]

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P.O. Sindri Institute
Dhanbad- 828123

First mid semester Examination -2023.
Branch- Electrical Engg., E.C.E., C.S.E. and Information Technology.
Subject- Physics.

Full Marks - 20.

Answer any **five** questions. All question carry equal marks.

Time- $1\frac{1}{2}$ Hours.

1. Answer all questions.

(I) The wave function for the motion of the particle in a one- dimensional potential box of lengths is given by $\psi_n = A \sin(\frac{n\pi x}{a})$. Where A is the normalization constant. The value of A is

1×4=4.

(a) $\frac{1}{a}$ (b) $\sqrt{\frac{2}{a}}$ (c) $\sqrt{\frac{a}{2}}$ (d) none of these.

[CO1]

Where a is the length of the box.

(II) In quantum mechanics a classical dynamical quantity is replaced by the corresponding quantum mechanical operator. Accordingly, the x component of momentum p_x is replaced by

(a) $i\hbar \frac{\partial}{\partial x}$ (b) $-i\hbar \frac{\partial}{\partial x}$ (c) $i\hbar \frac{\partial}{\partial x}$ (d) none of these.

[CO1]

(III) The density of a simple cubic crystal is given by (a) $\frac{3M}{a^3 N_A}$ (b) $\frac{2M}{a^3 N_A}$ (c) $\frac{M}{a^3 N_A}$ (d) none of these.

[CO2]

Where terms have their usual meanings.

(IV) The atomic radius in a F.C.C. cell is

[CO2]

(a) a (b) $\frac{a\sqrt{3}}{4}$ (c) $\frac{a}{2\sqrt{2}}$ (d) none of these.

Where terms have usual meaning.

OR

What do you mean by an operator? Find the operator corresponding to (a) Energy (b) linear momentum and (c) Kinetic energy.

1+3=4, [CO1]

2. (a) Derive the Schrodinger time dependent and time independent wave equation in one dimension. Write the Schrodinger time dependent and time independent wave equation in three dimensions.

(b) Explain the physical significance of the wave function. ✓

2+1+1=4, [CO1]

3. Write the Schrodinger equation in one dimension for the particle in an infinite deep potential well (rigid box) and solve it to obtain the Eigen values and Eigen function. Consider an electron in an infinite well of width $0.5 \text{ \AA}$. Calculate energy of it in ground state.

1+2+1=4, [CO1]

4. What are Miller indices? How they are determined? Show in a diagram the (111) plane of a cubic lattice.

1+2+1=4, [CO2]

5. (a) Discuss the classification of solids into metals, semiconductors and insulators, on the basis of band theory.

(b) The unit cell of copper is a cube. The side of the cube is $3.6 \text{ \AA}$. If the unit cell in copper Solid are lined up side by side, how many unit cells will be there along 10mm length of the solid.

3+1=4, [CO2]

6. What are density of energy states in metals? Derive an expression for density of energy states of a metal. 1+3=4, [CO2]

7. Explain the followings (a) Expectation values (b) Eigen values and Eigen functions.

2+2=4, [CO1]

B. I. T. Sindri,
Dhanbad-828123

B. Tech. 1st Semester 2nd Mid-Semester Examination-2023
Subject: Engineering Physics (BSC02)

Branch: Electrical Engg., E. C. E., C. S. E and Information Technology

Time Allowed: 1.5 Hours.

Total Marks: 20

Candidates are required to give their answers in their own words as far as practicable.

Answer any Five questions.

1. Choose the correct option

(1×4= 4)

(i) The Fermi level in a p-type semiconductor lies close to

(CO 3)

- (a) Top of valence band (b) Top of conduction band
(c) Bottom of conduction band (d) None of these

(ii) The direction of Hall voltage is

(CO 3)

- (a) Perpendicular to both applied electric & magnetic field (b) Parallel to applied electric field
(c) Perpendicular to applied magnetic field (d) None of these

(iii) What is the wavelength of the emitted laser by a carbon dioxide?

(CO 4)

- (a) 8.2 μm (b) 10.6 μm (c) 11.4 μm (d) 12.5 μm

(iv) The following is not a candidate material for laser source in Fiber optics

(CO 4)

- (a) Nd-YAG (b) He-Ne (c) Argon (d) Phosphorous

2. What is Hall effect. Show that the Hall coefficient is independent of the applied magnetic field and is inversely proportional to the current density and electric charge. Mention the important applications of Hall effect. (1+2+1) (CO 3)

3. What is Fermi Energy? Discuss the behaviour of Fermi level with variation of temperature. Evaluate the Fermi function for energy $K_B T$ above the Fermi energy. (1+2+1) (CO 3)

4. What is law of mass action? Drive an expression for electron and hole concentrations in an intrinsic semiconductor. (1+3) (CO 3)

5. (a) Write the abbreviation of laser. What are the essential requirements for laser action? Discuss the important features of stimulated emission of radiations. (1+1+1)(CO 4)

(b) The emission wavelength is 6000 Å and the lifetime is 10^{-6} s. Determine the coefficient for the stimulated emission. (1) (CO4)

6. Describe the construction and working of He-Ne laser. How it is superior to a ruby laser? (3+1) (CO 4)

7. Write short notes on

(2+1+1) (CO 4)

(i) Spontaneous and stimulated emission of radiation.

(ii) Population inversion and pumping

(iii) Optical resonator

B. I. T. SINDRI
DHANBAD-828123

B. Tech. 1st Semester Supplementary Mid-Semester Examination-2023
Subject: Engineering Physics (BSC02)

Branch: Electrical Engg., E. C. E., C. S. E and Information Technology

Time Allowed: 1.5 Hours.

Total Marks: 20

Candidates are required to give their answers in their own words as far as practicable.

Answer any **Five** questions.

1. Choose the correct option

(1×4= 4)

(i) Fermi-Dirac distribution function is given as

(CO 2)

(a) $F(E) = \frac{1}{1 + e^{(E-E_F)/KT}}$

(b) $F(E) = \frac{1}{1 - e^{(E-E_F)/KT}}$

(c) $F(E) = 1 - \frac{1}{1 - e^{(E-E_F)/KT}}$

(d) None of these

(ii) For elements having energy gap more than 5eV, act as:

(CO 2)

(a) Semiconductors

(b) Insulators

(c) Conductors

(d) Superconductors

(iii) If N_1 and N_2 are the number of atoms in ground and excited state, respectively, then in population inversion

(CO 4)

(a) $N_1 < N_2$

(b) $N_1 > N_2$

(c) $N_1 = N_2$

(d) None of these

(iv) Ruby laser is a

(CO 4)

(a) Four-level laser

(b) Three-level laser

(c) Two-level laser

(d) None of these

2. What are lattice parameters? Discuss the seven types of crystal systems on the basis of lattice parameters.

(1+3) (CO 2)

3. What is interplanar spacing? Derive the formula for calculating interplanar spacing.

(1+3) (CO 2)

4. What do you mean by density of states in metals? Derive an expression for the density of energy states in metal.

(1+3) (CO 2)

5. What do you understand by population inversion? Discuss the terms spontaneous and stimulated emission of radiation and find the relation between their coefficients.

(1+3)(CO 4)

6. Describe the construction and working of ruby laser.

(4) (CO 4)

7. Write short notes on

(1+2+1) (CO 4)

(i) Interaction of radiation with matter.

(ii) Numerical aperture & acceptance angle

(iii) Condition for light amplification

MID SEMESTER – I EXAMINATION : 2023**BIT, SINDRI (B.TECH/ 1st SEMESTER)**Jharkhand University of Technology

(For Electrical, ECE, CSE & IT Branches only)

Full Marks : 20

Time : 1 ½ Hrs

Answer any FIVE out of the SEVEN questions as provided hereinunder. Try to write clearly in your own language as far as possible. Each question carries equal weightage of Marks.

1. What is the significance of introducing 'Indian Knowledge System' as a subject of study for Engineering Students under NEP 2020? CO1, CO2
2. Why do you think that India inspite of having Rich Ancestral Knowledge System and scientific legacy does not find much mention in Global Knowledge Base and Education System? CO3
3. What are the different Schools of Art that had flourished in Ancient India? Explain in details the period, main features of these Schools of Art and the places of their origins? CO1, CO3
4. Write short notes on any two of the following : CO1, CO2, CO4
 - (a) Patanjali
 - (b) Yoga in Ancient India
 - (c) Pottery and Utensils in Ancient India
5. Discuss the level of knowledge, expertise in the field of Astronomy and Mathematics in Ancient India. Do you think Indian Ancestral Astronomical and Mathematical knowledge/ work have contributed anything to the world of Science? CO1, CO2, CO6, CO7
6. How do you see Ayurveda as an alternate system of medicine? Elucidate in details the rich legacy of knowledge that India inherited in the field of Ayurveda from our ancestors in the form of seers, sages and saints. CO1, CO2
7. Critically examine the rich ocean of Ancestral Indian Knowledge System vis-a-vis its relevance to present day science and technology while focussing on its usefulness to mankind. CO1, CO6, CO7

BIT, SINDRI

JHARKHAND UNIVERSITY OF TECHNOLOGY

MID SEMESTER – II EXAMINATION (B.TECH, 1ST SEM) : 2023-24

(For Electrical, ECE, CSE & IT Branches only)

Full Marks : 70

Time : 3 Hrs

Answer any FIVE out of the SEVEN questions as provided hereinafter. Question 1 is COMPULSORY. Try to write clearly in your own language as far as possible. Each question carries equal weightage of Marks.

1. Answer the following questions as directed :

- Kautilya was also known as Vishnugupta. (State True or False)
- Vagbhata was the author of two great treatises of Ayurveda namely _____ and _____. (Fill in the blanks)
- _____ in 800 BCE calculated the value of 'pi'. (Fill in the blank)
- Brihadisvara Temple is located in _____. (Fill in the blank)
- It is said that in Ancient India, _____ Maharshi invented an Electric-Cell made of an Earthen Pot. (Fill in the blank)
- Sanchi Stupa was commissioned by _____ in 3rd century BC. (Fill in the blank)
- The binary number system was first described by the Vedic scholar _____. (Fill in the blank)
- The Indian Knowledge Systems comprise of Jnan, Vignan and _____. (Fill in the blank)
- _____ was a Sanskrit grammarian who gave a comprehensive and scientific theory of Phonetics, Phonology and Morphology. (Fill in the blank)
- The origin of pottery in India dates back to _____ and _____ periods. (Fill in the blanks)
- The Ancient Indus Valley Civilization City Mohenjo Daro is also known as _____. (Fill in the blank)
- The three main styles of ancient Indian temple architecture are Nagara, _____ and _____. (Fill in the blanks)
- Vedic Mathematics consists of _____ sutras called formulas and _____ sub-sutras called sub formulas. (Fill in the blank)
- The Palas developed Buddhist centres of learning such as the _____ and the _____ universities. (Fill in the blank)

2. "We owe a lot to the ancient Indians, teaching us how to count. Without which most modern scientific discoveries would have been impossible." Justify this statement of Albert Einstein in the light of rich 'Indian Knowledge System' that has proved to be a real asset for entire mankind.

3. Do you agree that Acharya Vagbhata, the son of Simhagupta was the one who first presented Ayurveda in a most simple and comprehensive way? If so, then explain in details the contribution of Acharya Vagbhata in the field of Ayurveda. How do you see Ayurveda as an alternate system of medicine and do you think that it is posing a great deal of challenge to other branches of modern medicine?

4. Explain the term 'Sustainable Environment'? Do you think that 'Sustainable Environment' is the need of the hour in the light of new environmental challenges of Global Warming and depleting natural resources that have posed a great deal of threat to the very survival and existence of Human Beings?

5. Which Ancient Indian Sage had compiled and gave us a comprehensive text on Yoga called Yoga Sutras? Describe in detail the different elements of practice of Yoga while discussing the significance and benefits of all the elements of Yoga Sutras.

6. Sir Isaac Newton is credited for giving us the concept of 'Law of Gravity'. Do you think the 'Law of Gravity' has got any ancient India connection? If yes, explain how and why?

7. Write Short notes on :

- Drainage System and Township Planning of Harappan Civilization
- Minting and Casting of Metals in Ancient India
- Medicinal use of Forest Resources
- Role of Aryabhatta in Indian Knowledge System

BIT SINDRI DHANBAD (JHARKHAND) – 828 123
Mid Semester Exam - II (MECH & PRODUCTION - 2024)
1st Semester, Section - A & B
SUBJECT: ENGINEERING MECHANICS

Time Allowed: 90 Minutes

Full Marks: 20

INSTRUCTIONS

1. There are 06 questions answer any 04 questions. Each question carries equal marks.
2. Notations are their usual meaning.
3. Assume suitable data for any missing data.

- Q.1** a) Write the scalar components of the acceleration and also show its directions, if a particle moving with velocity, v on a curved path when radius of curvature is ρ . [2]
- b) The position of a particle which moves along a straight line is defined by the relation, $x = t^3 - 6t^2 - 15t + 40$, where x is expressed in meters and t in seconds. Determine:
- i) the time at which the velocity will be zero,
 - ii) the distance traveled by the particle at that time,
 - iii) the distance traveled by the particle from $t = 4$ second to $t = 6$ second. [3]
- Q.2** a) Write the expression for distance travelled in n^{th} second of a body is having linear motion. [2]
- b) A ball is tossed with a velocity of 10 m/s directly vertically upward from a window located 20 m above the ground. Knowing that the acceleration of the ball is constant and equal to 9.81 m/s^2 downward. Determine:
- i) the velocity v and elevation y of the ball above the ground at any time t ,
 - ii) the highest elevation reached by the ball and corresponding value of t ,
 - iii) the time when the ball will hit the ground and the corresponding velocity. [3]
- Draw the $v - t$ and $y - t$ curves.
- Q.3** a) Define linear momentum and linear impulse of a particle of mass, m acted upon by a force, F . [2]
- b) A 10 kg package drops from a downhill into a 25 kg cart with a velocity of 3 m/s. Knowing that the cart is initially at rest and roll freely. Determine:
- i) the final velocity of the cart,
 - ii) the impulse exerted by the cart on the package,
 - iii) the fraction of the initial energy lost in the impact. [3]
- Q.4** Car A and Car B are moving on a road with different velocities is shown in Fig. 1. The velocity and tangential acceleration of car A are $v_A = 10 \text{ m/s}$ and $(a_t)_A = 5 \text{ m/s}^2$ respectively whereas velocity and acceleration of car B are given by, $v_B = 18.5 \text{ m/s}$ and $a_B = 2.0 \text{ m/s}^2$. Find: a) $v_{A/B}$ and b) $a_{A/B}$ [5]

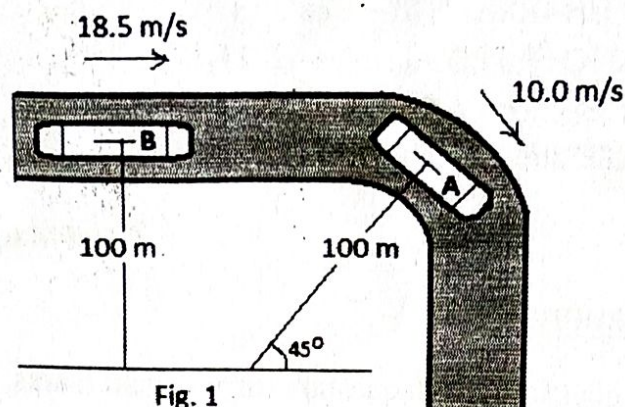


Fig. 1

- Q.5** A 1000 kg car starts from rest at point 1 and moves without friction down the track shown in Fig. 2. Determine:
- the force exerted by the track on the car at point 2, where the radius of curvature of the track is 6.0 m .
 - the minimum safe value of the radius of curvature at point 3.
- [5]

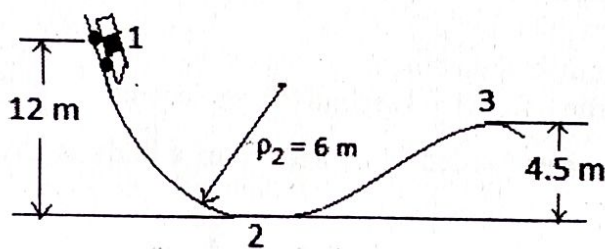


Fig. 2

- Q.6** Two blocks A and B of weights 1.0 kN and 2.0 kN respectively are in equilibrium positions as shown in Fig. 3. If the coefficient of friction between the two blocks as well as the block B and the floor is 0.3 , find the force, P required to move the block B.
- [5]

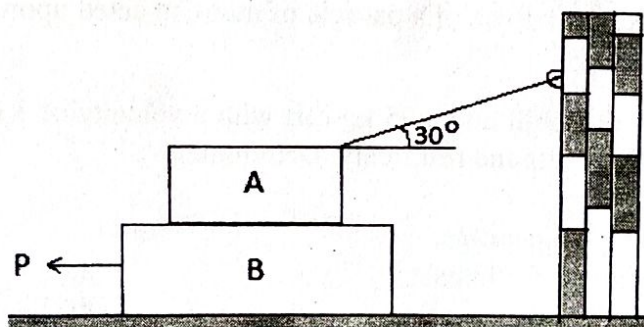


Fig. 3

Jharkhand University of Technology, Ranchi

BIT SINDRI, DHANBAD

B.Tech. 1st Semester

3rd Mid Semester Examination 2023

Subject: Engineering Mechanics

Time: 1:30 hrs

Full Marks: 20

Instructions:

01. Candidates are required to give the answers in their own words as far as practicable.
02. Answer any **FIVE** Questions.
03. All questions carry equal marks.

1. Figure 1 shows two vertical forces and a couple of moment 2000 N-m acting on horizontal rod which is fixed at end A.

- (i) Determine the resultant force of external forces,
- (ii) Distance of resultant force from point A,
- (ii) Determine an equivalent system through A.

(1 + 2 + 1 = 4 marks)

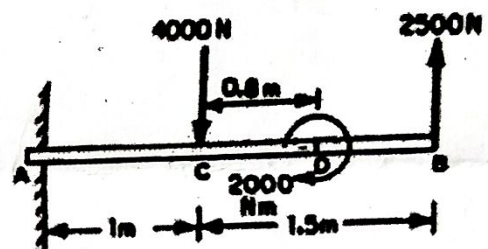


Figure-1

- ✓ 2. Two smooth circular cylinders, each of weight $W = 1000\text{ N}$ and radius 15 cm, are connected at their centres by a string AB of length = 40 cm and rest upon a horizontal plane, supporting above them a third cylinder of weight = 2000 N and radius 15 cm as shown in Figure 2. (i) Draw the free body diagram of each cylinder (ii) Find reactions at the points of contact F and G only.

(2 + 2 = 4 marks)

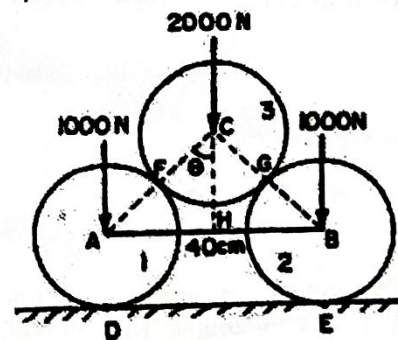


Figure-2

3. Find the expression of tensions ratio (T_1/T_2) for flat belt in terms of co-efficient of friction (μ) and angle of lap (θ). (4 marks)

- ✓ 4. Two blocks A and B, connected by a horizontal rod and frictionless hinges are supported on two rough planes as shown in Figure 3. The coefficients of friction are 0.3 between block A and the horizontal surface, and 0.4 between block B and the inclined surface. If the block B weighs 100 N. (i) Draw free body diagram of block A and block B (ii) what is the smallest weight of block A, that will hold the system in equilibrium?

(2 + 2 = 4 marks)

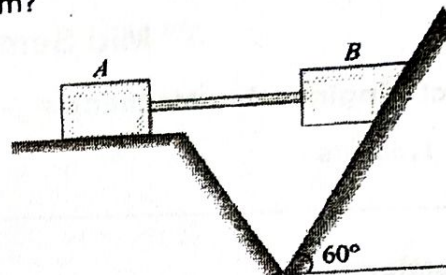


Figure-3

- ✓ 5. For a given Truss as shown in Figure 4. Find (i) the reactions at point C and E; and (ii) internal force in DE using section method.

(2 + 2 = 4 marks)

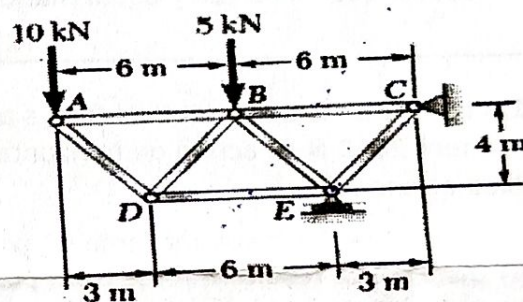


Figure-4

6. Refer Figure 5, A hammer of mass 0.5 kg hits a nail of 25 g with a velocity of 5 m/s and drives it into a fixed wooden block by 25 mm. Find the resistance offered by the wooden block.

(4 marks)

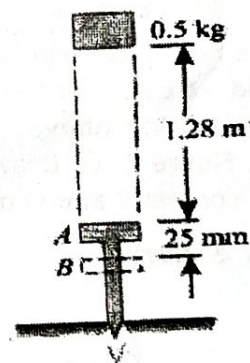


Figure-5

- ✓ 7. Write short notes on any two of the following:

- Why couple is a Free Vector?
- Work-Energy Principle
- Assumptions in Trusses
- Impulse-Momentum Principle.

(2 x 2 = 4 marks)